

From Trajectories to Evidence: Auditing Observation Loss and Identifiability in Computer-Use Agent Evaluation

Draft v0.3 — authors unset — venue unset

13 September 2026

Abstract

Computer-use-agent reliability scores are evaluation outputs, not direct observations of episodes. That fact is established background: validity theory already treats scores as interpretations requiring support, and recent CUA audits already treat a score as the output of a pipeline. This paper instead freezes an evaluation instrument and treats the instrument as the experimental object. In a locked trajectory-to-evidence transformation, 39 of 59 rows with recoverable gold were missed, including 13 cases in which gold had already entered the extractor’s accumulator and was then discarded. A pre-specified permissive repair recovered eight additional correct matches but released eighteen wrong ones; composing all repairs was worse than leaving the extractor frozen. Constructing an observation-grounded last-text measurement then exposed further boundaries: 40 completed episodes with 30 abstentions; unstructured last-text coverage 0.1667; interface Form 0.9333 with two-sided correspondence unevaluable; and a competent sample of 30 HIT and 0 MISS ($I_{CC} = 0$). We do not propose a replacement reliability metric. A reliability number in this frozen setting should be accompanied by its observation channel, evidence rule, abstention policy, non-leakage constraint, and identifiability status before it is interpreted as a reliability claim.

1 Introduction

A computer-use-agent (CUA) reliability score is an evaluation output, not a direct observation of the episode. That fact is established background, not this paper’s contribution. Leaderboards compress trajectories into binary outcomes or scalars [Jang et al., 2026, Xie et al., 2024, Zhou et al., 2024]. The scalar is produced by a pipeline

$$\tau \longrightarrow I \longrightarrow P \longrightarrow E \longrightarrow Y \longrightarrow S \tag{1}$$

(Figure 1): a declared observation channel I , a parser P , an evidence classification E , a correspondence decision Y , and a reported score or selection S . Dong et al. already treat a CUA score as the output of such a pipeline and show that public FAIL verdicts can be wrong [Dong et al., 2026]. Validity theory already treats scores as interpretations and uses that require supporting argument [Kane, 1992, 2013, Messick, 1995].

Existing work has documented evaluator error, protocol shortcuts, last-frame observation loss, and construct-validity failures in benchmarks [Dong et al., 2026, Shao et al., 2026, Xue et al., 2025, Rosset et al., 2026, Bean et al., 2025]. The remaining question is narrower. What happens when the evaluation *instrument* itself is frozen and treated as the experimental object—not the agent, and not the published verdict?

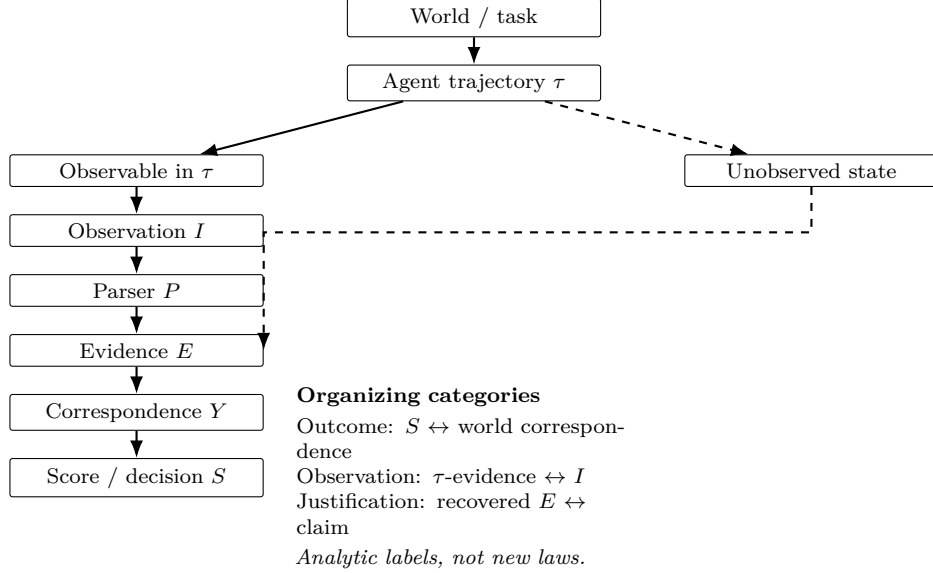


Figure 1: A reliability score is constructed along $\tau \rightarrow I \rightarrow P \rightarrow E \rightarrow Y \rightarrow S$. Unobserved world state may be the intended measurand; it is not automatically in I . The three gaps are organizing categories for the evidence below, not a claim that this paper discovered three fundamental laws of measurement.

We show four things in one frozen CUA evaluation setting. Recoverable typed evidence can be discarded after the extractor has already collected it. Making that extractor more permissive need not restore the intended measurement: in this instrument a plurality repair released more wrong matches than correct ones. Constructing an observation-grounded last-text measurement then exposes further boundaries: completion without determining evidence, interface compliance without two-sided identifiability, and competence without negative variation. A reliability number should therefore be accompanied by an explicit evidential basis—observation channel, evidence rule, abstention, non-leakage, and identifiability—before it is interpreted as a reliability claim.

We do not claim a first measurement framework for agents, a first demonstration that scores are constructed, or that existing CUA benchmarks are invalid. We do not propose a replacement reliability metric. The object of evaluation changes from “which agent scores highest” to “what this frozen protocol can actually justify.”

2 Background and related work

2.1 CUA and agent evaluation

Desktop and web CUAs are scored in OSWorld, WebArena, VisualWebArena, and MyPCBench [Xie et al., 2024, Zhou et al., 2024, Koh et al., 2024, Jang et al., 2026]. WebArena-Verified re-audits web tasks toward deterministic, offline evaluators [El Hattami et al., 2025]. Tool-using agents are scored by database-state match and multi-trial reliability [Yao et al., 2025]. AppWorld grades interactive coding by state-based unit tests, including collateral damage [Trivedi et al., 2024]. LLM-as-judge protocols and their prompt sensitivity are documented for chat evaluation [Zheng et al., 2023, Bellibatlu et al., 2026].

Shao et al. show that agent-benchmark scores support capability claims only when the protocol

keeps the intended capability necessary for success; they audit 2,385 traces across 15 benchmarks for exposure, exploitation, and score inflation [Shao et al., 2026]. That is protocol validity in the sense of shortcuts and invalid scoring paths. This paper does not rest on the claim that “scores justify claims,” which Shao already makes. It asks a different grain: once a protocol is frozen, what evidence does its observation channel recover, and can that evidence identify the correspondence claim one wanted to make?

2.2 Validity and measurement

We do not invent validity theory. Messick treats validity as scientific inquiry into score meaning and use [Messick, 1995]. Cronbach and Meehl locate construct validity in the network of inferences around a test [Cronbach and Meehl, 1955]. Kane’s argument-based approach is the principle we rely on: it is the proposed interpretations and uses of scores that are validated, not the scores themselves; more ambitious claims require more support; validating an interpretation does not automatically validate a use [Kane, 1992, 2013]. Jacobs and Wallach import measurement modeling into computational systems: unobservable constructs are operationalized, and harms follow from mismatches between construct and operationalization [Jacobs and Wallach, 2021]. Raji et al. and Bowman and Dahl diagnose construct-validity and benchmarking failures when a finite task is treated as a general capability measure [Raji et al., 2021, Bowman and Dahl, 2021]. Bean et al. review 445 LLM benchmarks and give an operational construct-validity checklist [Bean et al., 2025].

What this paper adds is not a new validity framework. It is a CUA-specific operationalization at the trajectory–observation–evidence grain, under a *frozen* observation protocol, with an empirical isolation of post-collection discard, a signed repair experiment, explicit abstention as missing correspondence, and a constructive demonstration that a two-sided claim can remain unidentifiable under competent execution. Notation for that operationalization is bookkeeping, not novelty.

2.3 Evaluator and observation audits

Dong et al. is the closest CUA paper and must be confronted directly [Dong et al., 2026]. They state that a score is the output of a pipeline in which tasks can be stale, trajectories can omit decisive visual evidence, evaluators can reject valid alternatives, and aggregate reports can hide the cause of failure. They organize reliability across task construction, trajectory observation, scoring, and reporting. They then retrospectively audit 150 public *failure-scored* trajectories from five benchmarks and find that 15.3% of FAIL verdicts are wrong (10.7% evaluator false negatives; 4.7% broken tasks), with a further 3.3% unclear from the released evidence. That is a verdict audit: is the published FAIL correct?

Xue et al. (Online-Mind2Web / WebJudge) show that last-frame and underspecified LLM judges mis-score web agents and propose preserving critical intermediate screenshots [Xue et al., 2025]. Rosset et al. argue that last-frame and overloaded full-trajectory judges both fail, and build a verifier that selects diagnostic screenshots per rubric criterion [Rosset et al., 2026]. Those papers establish last-frame observation loss as a known failure of *visual* CUA judging. They do not freeze a last-text typed extractor and measure discard after gold has already entered an accumulator.

WeaveBench reports that outcome-only grading overestimates pass rate relative to a trajectory-aware judge [Li et al., 2026]. Lù et al. make automatic trajectory judges themselves the object of evaluation [Lù et al., 2025]. Cao et al. document procedure-aware “corrupt success”; Advani documents false success as confident closure without the implied outcome [Cao et al., 2026, Advani, 2026]. Turk shows, in clinical LLM/agent evaluation, that a coverage score and a purpose-built tracking score can rank models differently under paired interventions [Turk, 2026].

2.4 Positioning of this work

This is not “just benchmark auditing,” but the distinction is not unprecedented in spirit. Dong audits *verdicts* on public traces. We audit a *frozen transformation* $I \rightarrow P \rightarrow E$ on trajectories we already hold, then try to *construct* an observation-grounded measurement and record where construction stops.

Dong: retrospective verdict audit over public CUA traces, focused on whether released FAIL verdicts are correct.

This paper: treats the evaluation instrument as the experimental object, freezes the observation/evidence transformation, audits evidence loss inside that transformation (including gold collected then discarded), experimentally tests permissive repairs, and records identifiability boundaries encountered when constructing an observation-grounded last-text measurement.

Shao’s axis is exposure $\rightarrow$ exploitation $\rightarrow$ misleading shortcuts. Ours is observation $\rightarrow$ evidence $\rightarrow$ correspondence, including measurement loss after evidence has become available. Neither axis subsumes the other. We do not claim that treating evaluation as measurement is new; we claim a concrete instrument-level result in a frozen CUA setting.

3 Evaluation as an observation-to-claim pipeline

Equation (1) is the object of analysis, not a newly discovered law. I is the declared observation channel (in the constructive studies: last assistant text, never screenshots or tool traces as a silent fallback). P recovers at most one unique candidate span, or none. E is DETERMINING iff that span parses as specification-kind k , not iff it equals $\text{value}(L)$. Y is HIT/MISS only when E is DETERMINING and L is independent of this episode’s answer. S is any scalar or selection computed from Y , or from a coarser rubric that never exposes Y .

Three analytic categories organize the evidence. They are not proposed as a new ontology, and they overlap Dong’s construction / observation / scoring / reporting stages [Dong et al., 2026]. We use them because they match the grain of this programme:

- **Outcome gap:** S need not correspond to independently measured world tracking.
- **Observation gap:** evidence in τ need not enter I or survive P .
- **Justification gap:** recovered evidence need not identify the reliability claim one wanted to make (including two-sided correspondence).

Where “measurement gap” would be read as a contribution of measurement theory, we mean only this local evaluation-measurement gap: a mismatch among score, observation, and justified claim under a declared protocol.

Two distinctions are load-bearing and are interpretive principles, not novel theorems.

Justification is not negation.

$$\neg \text{Justifiable}(C \mid \tau, I) \neq \text{False}(C).$$

If the pipeline cannot recover the evidence a claim requires, the claim is unjustified *from this measurement*. That is not a demonstration that the agent failed, cheated, or “made no claim.” The inequality is the ordinary consequence of separating evidential justification from truth.

Table 1: Study 1 primary lanes (existence sample). Invariance is Type A over tracking-valid pairs. Intervals are Clopper–Pearson 95% and are wide. Source: Stage-4 snapshot b7b4203.

Lane	Valid	Track-valid	Type A	Sens.	Type B
Claude	9	7	6	1	2
GPT-5.5	8	7	3	4	1
Qwen3.5-35B-A3B	1	1	1	0	0

What is in the trajectory is not what the instrument sees. Write $\text{Observable}_\tau(\tau)$ for evidence present in communication, and $\text{Observable}_I(\tau)$ for what the frozen interface extracts. These sets need not coincide. A typed fragment that exists in τ but is not recovered by I/P is *measurement loss*, not absence of a claim.

We write

$$\mathcal{M}(\tau, \mathcal{I}) = \{ C : \text{the evidence } C \text{ requires is recoverable under } I \text{ and the inference is non-circular} \} \quad (2)$$

as notation for the set of claims justified under a declared observation protocol. It is not a reliability metric $R(\tau)$, not a validity theorem, and not an empirical implementation (none was run).

4 Study 1: Score–outcome dissociation

Can movement or stability of an evaluation score be interpreted as corresponding change in agent–world behavior?

This study is an existence / dissociation demonstration, not a prevalence estimate. We hold instruction, interface, and judge fixed, and apply a locked intervention to a pre-registered determining set in a seeded MyPCBench guest [Jang et al., 2026]. Tracking is a typed match of the final answer against guest gold, with no judge in the loop. Score sensitivity is exact equality $S_{\text{CF}} = S_{\text{Base}}$ versus any nonzero delta. A pair is Type A if tracking holds and the score is invariant; score-sensitive if tracking holds and the score moves; Type B if tracking is incomplete while the score is high and unchanged. A pair is valid only if both legs terminate canonically. Execution failures are absent observations, not tracking misses.

The frozen schedule executed 46 of 50 intended cells (92 legs). 35 legs are execution failures; 24 pairs are valid. Primary lanes are reported separately (Table 1). A size ablation and an exploratory Flash lane exist and are not pooled. We do not headline a pooled invariance rate, and we do not treat 24 pairs as an estimate of how often scores fail to track the world.

Claude: invariance $6/7 = 0.857$ ([0.421, 0.996]). GPT-5.5: $3/7 = 0.429$ ([0.099, 0.816]). Qwen3.5-35B-A3B: one valid pair, Type A ([0.025, 1.000]). The licensed claim is the phenomenon: in this sample the reported score can remain unchanged across materially different outcomes, and can vary without a corresponding change in tracking. A stable score does not certify tracking; a moving score does not certify a tracking failure. Paired world interventions follow a counterfactual design [Pearl, 2009]; they are a method for this dissociation, not the paper’s main contribution.

5 Study 2: Selection and coverage

Even when the evaluator is operationally well-defined, evaluation and selection can change the evidential basis of a reliability comparison. That is the locked heading of this study. It is not a claim that “the evaluator changes the truth.”

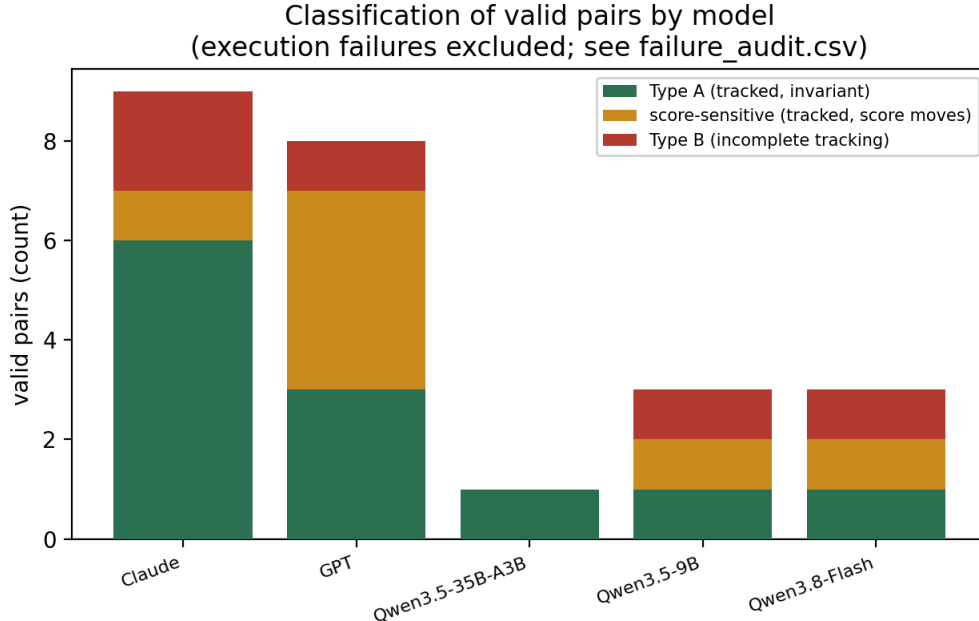


Figure 2: Study 1 protocol check: the intervention moves guest gold on the determining set. Tracking is coded from the final answer against that gold, never from S .

Table 2: Coverage under the frozen instrument. A valid pair requires both legs to terminate. Claude’s mean S^0 is a single pair. Freeze: tag `paper2-frozen` on commit 39cc662. Not a ranking of 57 tasks.

Agent	DONE/57	$ \mathcal{A} $	mean S^0	mean STS
GPT-5.5	32	9	69.3	0.130
Qwen3.8-Flash	29	8	95.8	0.229
Claude Opus 4.6	4	1	100	0

Three agents each ran a pre-registered schedule of 57 legs: GPT-5.5, Qwen3.8-Flash, and Claude Opus 4.6 (Table 2). The inclusion threshold $n_{\min} = 3$ valid pairs to be ranked, and the rule that fewer than three ranked agents means the confirmatory selection test is *not evaluated*, were committed before any outcome existed.

Claude terminated on four legs across three tasks, so $|\mathcal{A}| = 1$. The ranked roster is two agents, which is fewer than three, and the confirmatory selection test is therefore not evaluated—a stated outcome of the plan, not a failed experiment. We do not compare models on 57 tasks, and we do not treat $|\mathcal{A}|$ as a prevalence of selection bias.

Binary track $Y = 1$ requires every positive-weight component of the determining set to match on both legs. $Y = 0$ on all 18 valid pairs. There is no positive class, so S^0 cannot be calibrated to $P(Y = 1)$ on this universe.

On a four-task common support, labelled exploratory, a paired bootstrap (seed 20260904, $B = 5000$) gives Flash-minus-GPT $\Delta S^0 = +46.5$ (95% CI [21.0, 72.0]) and $\Delta \text{STS} = -0.042$ ([−0.125, 0.000]). The reversal is strict on one task (`retrieval-f009`) and tied on three. Removing that task makes the STS comparison a tie. We make no claim about the prevalence of ranking instability.

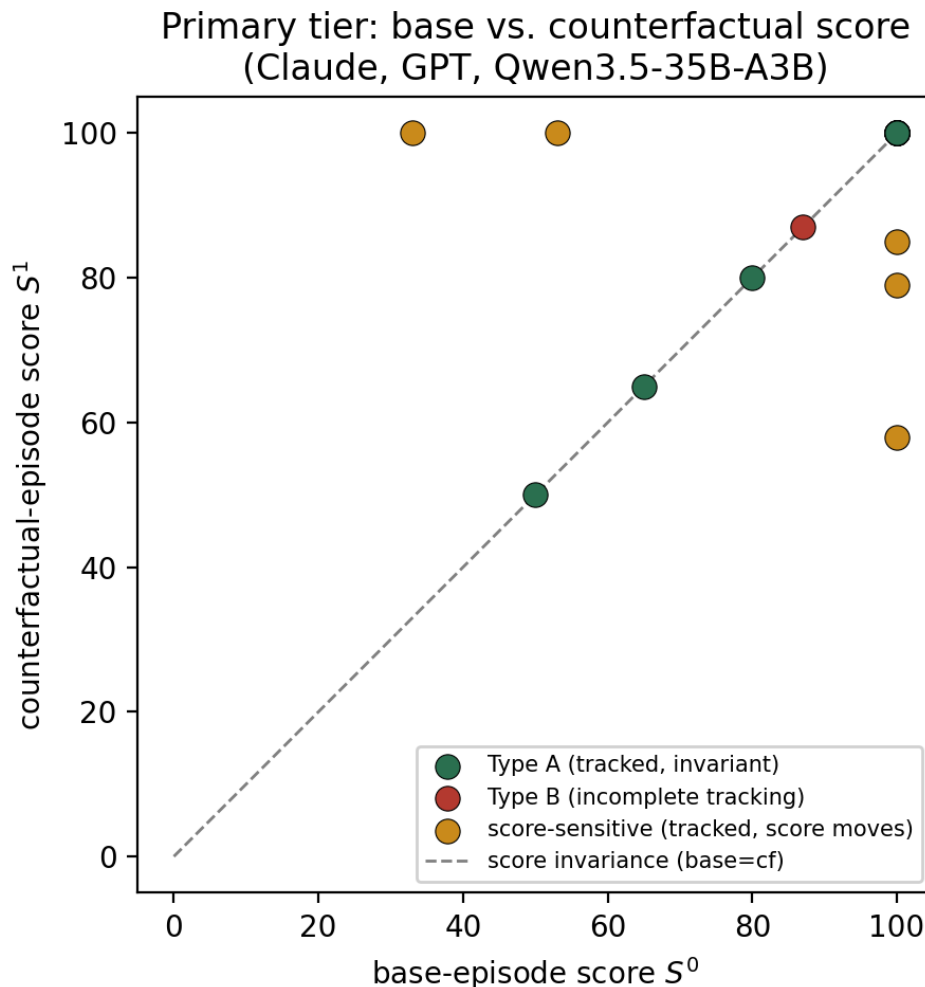


Figure 3: Study 1 primary lanes: paired base vs. counterfactual scores on valid pairs. Ablation and exploratory lanes are not pooled into Table 1.

Conditioning on termination also changes what the score can see. Excluded cells score lower on average (mean S off $\mathcal{A}$: 46.4, 44.9, 26.1 against 69.3, 95.8, 100 on $\mathcal{A}$). What survives that average is sharp: 2, 2, and 1 excluded cells scored ≥ 90 with no canonical **DONE**. A completion-conditioned protocol cannot express rubric-visible progress without closure.

The role of the study is eligibility: a reliability comparison can become conditional on what the evaluation pipeline makes scorable.

6 Study 3: Evidence loss and repair

Studies 1 and 2 locate an outcome / eligibility gap. They do not say where information leaves the pipeline. This section is the empirical core. We freeze the evaluator and inspect the maps that turn a trajectory into a reported value. The contribution is not that missing evidence matters in general—Dong, Xue, and Rosset already document observation and evaluator failures [Dong et al., 2026, Xue et al., 2025, Rosset et al., 2026]. The contribution is isolating a mechanism *inside a frozen*

Table 3: Cause of the 39 recall misses. Meanings are locked. Frozen extractor 3242c30.

Cause	n	Meaning
M1a	13	gold entered found and was discarded
M1b	7	candidates disagree; gold is not among them
M2	9	no label hit anywhere in the answer
M3	5	label hit; gold outside every $\pm$ window
M4	5	instrument reported a non-gold value

Table 4: Layer repairs as diagnostic interventions, 134 rows. Sensitivity is over 59 R1-positive rows. R-CMP is bit-identical to FROZEN.

Config	Sens./59	Abst./134	M4/reported	MATCH dest./20
FROZEN	20	89	25/45	0
R-AGG	28	63	43/71	0
R-SCOPE	13	56	65/78	14
R-CMP	20	89	25/45	0
R-CHAN	15	104	15/30	7
ALL	10	79	45/55	16

trajectory-to-evidence transformation and testing whether permissive repair restores the intended measurement.

The unit is a row: one component of the determining set on one leg (134 rows over 57 legs). A row is R1-positive if the gold value is literally present in the agent’s answer ($n = 59$: 20 MATCH + 39 RECALL_MISS). The remaining 61 rows are ABSENT. Rates below are conditional on these audited sets. They are not a false-negative rate of the agent, not an evaluator-disagreement rate, and not a screenshot-omission rate.

The frozen instrument recovers 20 of 59 R1-positive rows (sensitivity 0.339) and misses 39 (Table 3).

M1a is post-collection discard. In 13 of 39 misses the extractor had already accumulated the gold value and fail-closed aggregation discarded it. In 7 of those 13, gold was a strict majority of the accumulated candidates. This is not agent failure: the gold string was in the answer and in the accumulator. It is not generic missing data: the observation existed at an intermediate instrument state. It is not last-frame screenshot omission: the channel is text. It is not ordinary judge error: no LLM judge sits in this extractor. M1 is not exclusive to misses (25 of 61 ABSENT rows are also M1); M1a is the cause that names recoverable-but-discarded evidence. No single-cause story is licensed. A pre-registered secondary prediction that misses would concentrate on injected-world tasks *failed* (confidence intervals overlap; we do not replace it with a reversal).

Repairs test mechanisms, not a method. Four layer interventions were specified before the run (Table 4): R-AGG (plurality; targets M1a), R-SCOPE (whole-answer window; targets M3), R-CMP (containment; targets entity M1b), and R-CHAN (delete markup; targets scaffolding). ALL composes them. No configuration is offered as a replacement instrument.

No repair dominates. R-AGG is the only intervention that raises sensitivity ($20 \rightarrow 28$). It releases 26 abstentions: +8 correct and +18 wrong. Recovering more evidence is not equivalent to recovering the intended measurement. R-SCOPE converts 14 of 20 already-correct rows into

Table 5: Constructive measurement boundaries. These are not four failed metrics and not a ranking of agents.

Stage	What was constructed	What failed / stopped	Scientific boundary
P4-B	Last-text HIT/MISS/ABSTAIN on 40 confirmatory episodes	30/40 ABSTAIN (<code>no_anchor</code>); E3/E4 not evaluable	Completion $\neq$ determining evi- dence in I
P4-C	Unstructured last-text metric v1	Flash coverage 0.1667 (gate fail)	Natural last-text can be sparse under successful execution
P4-C2	Declared CLAIM : inter- face (metric v2)	Form 0.9333 PASS; H3 NOT_EVALUABLE	Interface compliance $\neq$ two-sided identifiability
P4-D	Controlled negative op- portunities, no MISS quota	Flash 30 HIT / 0 MISS; $I_{CC} = 0$	Competence can coexist with in- sufficient negative variation

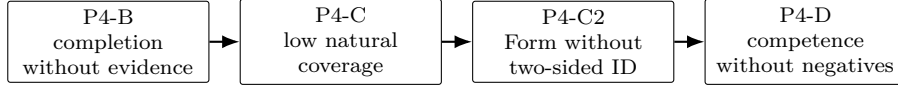


Figure 4: Constructive sequence: observation boundary $\rightarrow$ coverage boundary $\rightarrow$ interface boundary $\rightarrow$ identifiability boundary.

something else. R-CHAN destroys 7 MATCH rows. ALL is worse than the frozen extractor on sensitivity (10 vs. 20). A later-layer comparison repair cannot recover an upstream miss (R-CMP is bit-identical to FROZEN).

The licensed claim is not a theorem that parsers cannot be repaired. It is: permissive repair is not guaranteed to restore the intended measurement; in this instrument it released more wrong matches than correct ones, and composing all repairs was worse than leaving the extractor frozen.

On the same four-task support as Study 2, R-CHAN inverts the sign of Flash-minus-GPT Δ STS ($-0.0417 \rightarrow +0.1667$) because GPT collapses ($0.250 \rightarrow 0.042$) while Flash is unchanged. The effective number of independent task-clusters for the frozen Δ STS is one. This is an existence demonstration, not a claim that rankings are unstable in general.

A locked, corpus-independent reconstruction rule recovers 9 of 30 hand-written label components (G2 FAIL). Groundedness against the task definition is 100% (G1). A pre-specified comparative-scale gate required 20 independent task clusters; 16 of 28 candidates survived a frozen feasibility probe, so that experiment was not opened.

7 Constructive measurement boundaries

Having identified measurement loss on an existing instrument, we attempted to construct an observation-grounded measurement under controlled conditions. The channel is last assistant text. Execution status is never HIT/MISS. ABSTAIN is missing correspondence measurement, not task failure. The four stages below are a constructive sequence, not a leaderboard of failed metrics (Table 5, Figure 4).

7.1 P4-B: Completion without determining evidence

Confirmatory $N_B = 20$ clusters $\times$ two models = 40 episodes. All 40 legs terminated **DONE**. The instrument returned HIT 5, MISS 5, ABSTAIN 30. Every abstention is **no_anchor**: no instruction-anchor substring on a newline-terminated line of last text. The instrument did not then mine screenshots, tools, or intermediate turns. False-HIT and invariance gates passed (0 false HIT). Natural success and failure correspondence could not be fully evaluated because prespecified eligibility floors were not met (E3: 4 eligible clusters < 5 ; E4: 0 eligible). Status: positive-but-incomplete. Terminal completion is not observation-grounded measurement success.

7.2 P4-C: Natural last-text can be sparse

Metric v1 scores unstructured last text against independent gold. Flash confirmatory $N_C = 30$: HIT 3, MISS 2, ABSTAIN 25; coverage 0.1667; correspondence 0.6000 on the covered subset; abstention 0.8333. Coverage fails its gate (≥ 0.5). C1/C2 discrimination floors are not evaluable. Natural agents need not communicate determining evidence through an undeclared last-text channel. This is not a finding that the agent is incompetent.

7.3 P4-C2: Form without two-sided identifiability

Metric v2 requires a declared **CLAIM:** line (an interface constraint, not itself evidence). Flash $N = 30$: HIT 26, MISS 2, ABSTAIN 1, unscored 1; Form 0.9333 (gate pass); descriptive CC 0.9286. Ordinary MISS exists (two clusters). The C2-intended discrimination floor is **NOT_EVALUABLE** (8 eligible < 10 ; miss rate 0 on that slice). Interface compliance can pass while two-sided correspondence remains unevaluable. Form is not validity.

7.4 P4-D: Competence without negative variation

The question was observability of two-sided correspondence without a MISS quota. Flash $N = 30$: HIT 30, MISS 0, ABSTAIN 0; Form 1.0000; $I_{CC} = 0$ under the locked rule ($I_{CC} = 1$ iff at least 8 HIT and 8 MISS). Plus-competence held (G2: 10/10 HIT). Minus and plus-minus slices: 10/10 HIT each. Classification: W1—challenge weak / correspondence remains unobservable. First failing gate: G3. Even with controlled negative opportunities, this sample produced too few errors for the two-sided estimand to become identifiable. $I_{CC} = 0$ is an observability result, not a finding that the agent is unreliable, and not a claim that Flash is “100% reliable.” The workstream is closed: distractors are not amped, the parser is not retuned, and a MISS factory is not introduced.

8 Claim justification as a formal boundary

P4-M is theory-closed and empirically unimplemented. It is a claim-justification formalization: notation for which reliability claims remain sayable under a declared observation protocol. It is not a new theory of reliability, not a validity theorem, not a replacement metric, and not proven by P4-D.

The object is $\mathcal{M}(\tau, \mathcal{I})$ in Equation (2). The important distinction remains

$$\text{Observable}_{\tau}(\tau) \neq \text{Observable}_I(\tau).$$

An evidence fragment can exist in the trajectory and still not be recovered by a frozen instrument. That is measurement loss (Study 3; P4-B abstention), not “the agent made no claim.” **CLAIM:** syntax is an interface constraint on one particular I , not a definition of evidence.

E is kind-only: a span is DETERMINING if and only if it parses as $\text{kind}(L)$ from the task specification. 38.35 and 10.00 can both be DETERMINING for money. “Finished successfully” is NONDETERMINING; Y is undefined. E must not consume $\text{value}(L)$.

P4-D’s 30/30 HIT is closed evidence of unidentifiable two-sided correspondence. It is not a proof of $\mathcal{M}$.

8.1 Typed correspondence: a narrow no-leakage property

Under no-leakage assumptions (the maps I, P, E do not take $\text{value}(L)$ as input; L is locked from the spec/world, not from the evidence span; the match $\leftrightarrow_k$ is the pre-declared match on Val_k ; spans that require world-dependent denotation never become DETERMINING via this E),

$$Y = \text{HIT} \implies \text{parse}_k(s) = v \in \text{Val}_k \wedge v \leftrightarrow_k L.$$

This is a specification-level implication of the defined HIT relation. It is not a new theorem, not a reliability theorem, not a validity theorem, and not empirical validation of P4-M.

What is explicitly out of scope. Natural-language denotation; general reliability; agent competence; deployment safety; causal validity of the score as a treatment effect; completeness of $\mathcal{M}$; that a declared interface recovers English; that P4-M is “validated” by P4-D.

9 Unified discussion

The studies are not four independent papers. Table 6 is the fastest map of why they form one argument. Populations are not added. The three gaps remain analytic categories.

Outcome gap (Studies 1–2; supporting). Observed score $\neq$ underlying tracking or world state (Study 1 Type A/B on 24 valid pairs, existence only). Evaluation and selection further change the evidential basis of a comparison: coverage $|\mathcal{A}| = 9, 8, 1$; confirmatory selection not evaluated; binary $Y = 0$ on 18/18 valid pairs.

Observation gap (Study 3, P4-B/C; primary + constructive). Trajectory evidence $\not\subseteq$ evaluator observation. Study 3: 39/59 recoverable misses, 13 after gold had already been collected; R-AGG +8 correct / –18 wrong; ALL worse than frozen. P4-B: 40/40 DONE, 30 ABSTAIN. P4-C: unstructured last-text coverage 0.1667.

Justification gap (P4-C2/D; secondary). Observable evidence $\not\Rightarrow$ an arbitrary reliability claim. P4-C2: Form 0.9333 with two-sided discrimination unevaluable. P4-D: 30 HIT, 0 MISS, $I_{CC} = 0$ under competence. $\mathcal{M}(\tau, \mathcal{I})$ names the justified set as notation; NONDETERMINING text leaves Y undefined.

What the evidence does and does not establish. It establishes instrument-level measurement failure after evidence is available, non-dominance of permissive repair in this extractor, and identifiability failure for a two-sided estimand under a competent sample. It does not establish prevalence of M1a across CUA benchmarks, that last-text is the right observation channel for all CUA tasks, that public benchmarks are invalid, that Flash is unreliable, or that P4-M is empirically validated.

Table 6: Evidence for the central contribution. Populations are not added. This is not a prevalence table.

Source	Question	Frozen protocol	Interpretation	Does <i>not</i> establish
P3 M1a	Can gold be discarded after collection?	Extractor 3242c30	13/39 misses: gold in <code>found</code> , then dropped (7/13 majority)	First such phenomenon; agent FN rate
P3 R-AGG	Does permissive repair restore the measurement?	Pre-specified layer interventions	+8 correct / +18 wrong; ALL worse than frozen	Parsers cannot be repaired in general
P4-B	Does <code>DONE</code> imply determining E ?	Last-text I ; no screenshot fallback	40/40 <code>DONE</code> ; 30 <code>ABSTAIN</code>	Agents failed the tasks
P4-C	Is unstructured last-text dense?	Metric v1	Flash coverage 0.1667	Flash is incompetent
P4-C2	Does Form identify two-sided CC?	Declared CLAIM :	Form 0.9333; H3 N/E	Form is validity
P4-D	Is two-sided CC identifiable under competence?	No MISS quota	30 HIT / 0 MISS; $I_{CC} = 0$; G2 10/10	Flash is 100% reliable; metric is broken

10 A practical reporting framework

The eight slots below are a practical reporting checklist derived from the empirical and formal analysis. They are not a new metric, not a theorem, and not a validated universal standard. A benchmark author should be able to answer all eight before reporting a reliability number.

1. **Target.** Reliability of what system, on what task family, for what use?
2. **Measurand.** What independently defined state or outcome is correctness? Who locked L ?
3. **Observation channel.** What part of τ is actually observed (I)? What is forbidden as silent fallback?
4. **Evidence rule.** What makes an observation DETERMINING? What is NONDETERMINING?
5. **Correspondence.** How is evidence matched to the measurand? Is the matcher allowed to see $\text{value}(L)$?
6. **Abstention.** What happens when evidence is insufficient? Is `ABSTAIN` distinct from `MISS` and from execution failure?
7. **Non-leakage.** Can the procedure see the answer it is supposed to evaluate?
8. **Identifiability.** Does the sample contain enough positive *and* negative variation to support the intended claim (one-sided vs. two-sided)?

How this programme fills the slots, without pretending each slot is a new experiment: target is correspondence to an independently defined determining set, not vendor rank; measurand is guest gold / specification L ; channel is the audited MyPCBench extractor or frozen last assistant text; evidence is typed parse by $\text{kind}(L)$; correspondence is $\leftrightarrow_k$ disjoint from the judge in Studies 1–2; abstention is `ABSTAIN` / execution failure as missing observation; non-leakage is the P4 gold/ I split, and Study 3’s record of gold seen then discarded; identifiability is Study 2’s unevaluated selection test, P4-C2 H3, and P4-D $I_{CC} = 0$.

11 External transport boundary

We audited candidate public CUA trajectory sources against a frozen eligibility table (independent source; per-episode trajectory; final assistant natural-language message as I ; task spec; typed kind k ; independent L ; loss-versus-absence; offline replay; $N \geq 30$; outcome-blind sampling). No new agents, sites, docker, or parser were introduced. Phases 2–5 were not opened.

WebArena-Verified, as publicly released, is a task+evaluator specification (812 tasks) plus two demo logs whose `agent_response.json` is structured evaluator JSON, not a final assistant natural-language message. Official WebArena publishes task configs; published traces are Playwright/HTML/screenshot bundles, which the frozen I forbids as the observation channel. Named Hugging Face dumps inspected at card level, including later-appearing OSWorld success trajectories with screenshots and accessibility trees, were not established as satisfying typed k , independent L , and last-text I without an adapter. Using those dumps under the frozen contract would require a new parser/harness or a change of I , which the design forbids.

We did not identify a public corpus satisfying this specific frozen last-text typed observation contract. That is not a claim that existing CUA benchmark artifacts are generally not measurement-ready, not a ranking of WebArena agents, and not a claim that those benchmarks are invalid for task success. Transport analysis (HIT/MISS cells) was not opened. This STOP is the result.

12 Limitations

Typed candidate spaces only; natural-language denotation is out of scope. Last-text I is a severe observation channel; Xue and Rosset show that visual last-frame judging is itself lossy, so this paper’s channel is not proposed as canonical [Xue et al., 2025, Rosset et al., 2026]. External transport currently stops at eligibility. The reporting checklist is not a replacement for task-success evaluation, not a deployment-safety case, and not a proof that existing benchmarks are wrong. We do not claim that all reliability metrics suffer equally, or that scores are universally invalid. Studies 1–3 share a MyPCBench instrument family; P4 uses a constructed slate—populations are not added. Study 3’s ordering inversion has $n_{\text{eff}} = 1$. Study 1 and Study 2 are existence / eligibility demonstrations with small valid-pair counts and wide intervals. P4-M has no new empirical corpus. M1a is instrument-specific until transported; we did not transport it. Venue is unset.

13 Conclusion

We treated a frozen CUA evaluation instrument as the experimental object. Recoverable typed evidence can be lost after it has already been collected; making the extractor more permissive did not restore the intended measurement in this instrument; and constructing an observation-grounded last-text protocol exposed completion without evidence, Form without two-sided identifiability, and competence without negative variation. The practical consequence is a reporting discipline, not a new score: state the observation channel, the evidence rule, abstention, non-leakage, and identifiability before interpreting a reliability number. The central question is not “what is the reliability score?” but “what evidence does this frozen protocol observe, and what claim does that evidence justify?”

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